Brandolini and the Plan 3 Entry Theorem

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1 Purpose

This note rechecks whether the remaining entry theorem in the planar hybrid route can be supplied by Brandolini–Nitsch–Salani–Trombetti together with the other Plan 3 ingredients.

The answer is:

> *Brandolini supplies the correct Serrin-to-near-ball input, and the corrected $\sqrt{\delta_T}$-stopping scale makes this input usable. However, Brandolini does not by itself give the radial graph/radialization theorem needed for the sharp conclusion with constants independent of the stopped level. One planar no-fjord/radialization lemma is still missing.*

This is a positive recheck, not a negative one: the missing lemma is now much narrower than the previous graph-entry statement.

2 The stopped level

Let $n = 2$, normalize $|U| = \pi$, and set

$$\delta = \delta_T(U), \qquad a = \alpha(U).$$

In the bad alternative for the desired estimate, $a > M\sqrt{\delta}$. Choose

$$\lambda = K\sqrt{\delta}.$$

The volume-variable Chebyshev extraction, applied to the joint integrand $D_H + D_I$, gives a regular level $E_{\hat{t}} = \{u > \hat{t}\}$ with

$$|U \setminus E_{\hat{t}}| \leq \lambda, \qquad D_H(\hat{t}) + D_I(\hat{t}) \leq C\frac{\delta}{\lambda} = \frac{C}{K}\sqrt{\delta}. \tag{2.1}$$

The exact profile transfer gives

$$\delta_T(E_{\hat{t}}) \leq \frac{1}{1 - \lambda/\pi}\,\delta_T(U) \leq 2\delta \tag{2.2}$$

for small δ. This is the $u - \hat{t} = u_{E_{\hat{t}}}$ identity, not a monotonicity principle.

On the selected class, the D_H-to-L^∞ interpolation estimate from the companion note gives

$$\big\||\nabla u| - \bar{f}\big\|_{L^\infty(\partial E_{\hat{t}})} \leq C\left(\frac{\delta}{\lambda^2}\right)^{1/3} = CK^{-2/3}. \tag{2.3}$$

Thus K fixed and large makes $E_{\hat{t}}$ an approximate Serrin domain in the pointwise sense.

3 What Brandolini Gives

Let $v = u - \hat{t}$, so $v = u_{E_{\hat{t}}}$. After scaling v by the harmless dimensional factor that turns $-\Delta v = 1$ into Brandolini's $\Delta w = 2$ normalization, (2.3) implies

$$\big||Dw| - 1\big| \le \varepsilon_K \quad \text{on } \partial E_{\hat{t}}, \qquad \varepsilon_K \to 0 \quad (K \to \infty), \tag{3.1}$$

up to the harmless adjustment of the target constant coming from $|E_{\hat{t}}| = \pi + O(\sqrt{\delta})$.

Brandolini et al. give two different outputs.

3.1 Theorem 1: measure closeness to a union of balls

Their Theorem 1 needs smoothness, bounded diameter, the upper bound $|Dw| \le 1 + \varepsilon_K$, and $L^1(\partial E_{\hat{t}})$-closeness of $|Dw|$ to 1. Our L^∞ bound gives both boundary assumptions. The constant depends only on the diameter in this theorem. Hence $E_{\hat{t}}$ is close in measure to a finite union of almost unit balls:

$$\left| E_{\hat{t}} \Delta \bigcup_{i=1}^{m} B_i \right| \le C\,\varepsilon_K^{\beta}. \tag{3.2}$$

For K large, and after removing the satellite components controlled by D_I below, the volume normalization forces $m = 1$. Thus Brandolini gives a robust near-ball location and a center without importing BDV.

3.2 Theorem 2: annular containment

Their Theorem 2 says that if the domain is connected and $C^{2,\alpha}$, then

$$R_{\text{out}} - R_{\text{in}} \le C\,\varepsilon_K^{\mu}. \tag{3.3}$$

But here C depends on the $C^{2,\alpha}$ regularity constants of the boundary. For a stopped torsion level those constants are not uniform from the information currently proved. At depth $\lambda = K\sqrt{\delta}$, interior estimates only give schematically

$$\|D^2 u\|_{L^\infty} \lesssim \lambda^{-1}, \qquad \|D^3 u\|_{L^\infty} \lesssim \lambda^{-2}, \tag{3.4}$$

while $|\nabla u|$ is bounded below on the level. Thus a direct implicit-function $C^{2,\alpha}$ norm for $\partial E_{\hat{t}}$ can blow like a negative power of λ. Since K is fixed in the contradiction argument, ε_K is fixed; it cannot compensate for a constant blowing as $\delta \downarrow 0$.

Therefore Brandolini's Theorem 2 cannot be used as a black box with uniform constants unless one proves a new uniform regularity estimate for the stopped levels. That estimate would essentially be a graph-entry theorem in another form.

4 What Plan 3 Adds

The Plan 3 and Plan 2 files contain several ingredients that partly replace the regularity-dependent part of Brandolini.

4.1 Satellites

Let E_0 be the largest indecomposable component of $E_{\hat{t}}$. Since $D_I(\hat{t}) \le \eta$, where

$$\eta := CK^{-1}\sqrt{\delta},$$

the planar component estimate in `PLANAR_GRAPH_REDUCTION_PUSH.tex` gives

$$|E_{\hat{t}} \setminus E_0| \le C\eta^2 \le C_K \delta. \tag{4.1}$$

The Saint–Venant bound on each small component gives a negligible torsion contribution. Thus satellites are not an obstruction.

4.2 Closed holes

The same planar perimeter bookkeeping gives that bounded holes whose total boundary perimeter is p_h have total area at most $p_h^2/(4\pi)$. Since their perimeter is part of the isoperimetric excess at the good level,

$$|H_{\text{holes}}| \le C\eta^2 \le C_K\delta. \tag{4.2}$$

Filling closed holes is compatible with the torsion deficit by the monotone filling lemma in `PLANAR_RADIALIZATION_PROOF_ATTEMPT.tex`.

4.3 Outer annular control

Brandolini Theorem 1 gives small measure distance to one ball once K is large. Together with $D_I(\hat{t}) \to 0$, the planar Lemma 10 mechanism in Brandolini gives a small outer excursion. In dimension two the final step of that argument is purely length-theoretic: a connected set reaching a distance s outside the selected ball pays perimeter outside the ball at least comparable to s. Thus the outer radius can be made smaller than the fixed graph-entry threshold by choosing K large and then δ small.

This part does not require a uniform $C^{2,\alpha}$ constant in dimension two.

4.4 Outward fingers and bad rays

The $n = 2$ tentacle computation in the repo is favorable. A surviving tube of width w and length L has active-level cost

$$D_H \simeq L/w.$$

Thus a D_H-good level cannot carry an arbitrary long thin active finger. Plan 2's radial slicing gives a separate bad-ray multiplicity absorption once an annular bracket is known. These are genuine ingredients for converting annular near-ball information into radialization.

5 The Point Still Not Proved

The remaining issue is the inner and multiple-crossing geometry. Brandolini's Theorem 1 allows thin tentacles and gives only measure closeness. Brandolini's Theorem 2 would give inner and outer radii, but its constant depends on boundary regularity. Planar perimeter estimates rule out deep fjords at a fixed scale, and capacity rules out missing sets at positive torsion height, but the repo does not yet contain the arbitrary-geometry theorem that combines these facts at the sharp torsion scale.

The exact missing lemma is the following.

Conjecture 5.1 (Planar Brandolini–Plan 3 radialization lemma)**.** *Let $E = E_{\hat{t}} \subset \mathbb{R}^2$ be a regular $\sqrt{\delta}$-stopped level satisfying* (2.1)*,* (2.2)*, and* (2.3)*, with K fixed sufficiently large. Then, after translation and area normalization, there is a radial graph*

$$G = \{re^{i\theta} : 0 < r < 1 + \varphi(\theta)\}, \qquad \|\varphi\|_{L^\infty} \le \eta_0,$$

such that

$$|E\Delta G| \le C_K\sqrt{\delta}, \qquad \delta_T(G) \le C_K\delta. \tag{5.1}$$

This lemma is weaker than literal graph entry for E. It permits filling shallow fjords and holes and pruning outward debris. It is also exactly what the hybrid proof needs: the direct radial-graph theorem gives

$$\mathcal{A}(G)^2 \le C\,\delta_T(G) \le C_K\delta,$$

and then

$$\mathcal{A}(E) \le \mathcal{A}(G) + C|E\Delta G| \le C_K\sqrt{\delta}.$$

The transfer from E to U costs another $O(K\sqrt{\delta})$, so the desired estimate follows.

6 Verdict

The user's instinct is correct in the following precise sense. The repo contains the right ingredients around Brandolini:

$$\text{pointwise Serrin level} + \text{Brandolini measure near-ball}$$
$$+ D_I \text{ satellite/hole cleanup} + D_H \text{ thin-finger control} + \text{Plan 2 bad-ray slicing.}$$

These reduce the missing entry theorem to Conjecture 5.1.

But Conjecture 5.1 is not yet proved in the repo. The tempting shortcut through Brandolini's annulus theorem is not uniform, because its constant depends on $C^{2,\alpha}$ boundary regularity of the stopped level. The remaining work is therefore not another Serrin theorem. It is a planar geometric radialization theorem showing that the pieces not already handled by Brandolini measure closeness, D_I, D_H, and bad-ray slicing can be filled or pruned with $O(\sqrt{\delta})$ measure loss and $O(\delta)$ torsion-deficit cost.